

DANIEL LAHAV

CURRICULUM VITAE

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Department of Physics, Bar-Ilan University, Ramat-Gan, Israel | Citations: [ORCID](#), [Google Scholar](#)

EDUCATION

- Ph.D. in Physics, Bar-Ilan University, Ramat-Gan, Israel** 2024 – Present
Thesis: “Flexible Spintronics Under Strain”
- M.Sc. in Physics (Magna Cum Laude), Bar-Ilan University, Ramat-Gan, Israel** 2022 – 2024
Thesis: “Flexible Planar Hall Effect Sensors Under Strain”
- B.Sc. in Physics, Bar-Ilan University, Ramat-Gan, Israel** 2019 – 2022

LIST OF PUBLICATIONS

1. Nhalil, H., **Lahav, D.**, Schultz, M., Amrusi, S., Grosz, A., Klein, L., “Flexible planar Hall effect sensor with sub-200 pT resolution”, *Applied Physics Letters* 123(2), 2023. [doi:10.1063/5.0156588](https://doi.org/10.1063/5.0156588)
2. **Lahav, D.**, Schultz, M., Amrusi, S., Grosz, A., Klein, L., “Planar Hall Effect Magnetic Sensors with Extended Field Range”, *Sensors* 24, 4384 (2024). [doi:10.3390/s24134384](https://doi.org/10.3390/s24134384)
3. Vize, M., Alimi, R., **Lahav, D.**, Schultz, M., Amrusi, S., Grosz, A., Klein, L., “Magnetic Source Detection Using an Array of Planar Hall Effect Sensors and Machine Learning Algorithms”, *Applied Sciences* 15(2), 964 (2025). [doi:10.3390/app15020964](https://doi.org/10.3390/app15020964)
4. **Lahav, D.**, Schultz, M., Grosz, A., Klein, L., “Enhanced Magnetic Resolution in Elliptical Planar Hall Effect Sensors via Non-Collinear Anisotropy Engineering”, *Measurement* (under review).
5. Vize, M., Klein, L., Schneider, N., Fisher, E., **Lahav, D.**, Grosz, A., Alimi, R., “Enhancing Super-Resolution Fidelity through Physics-Guided Multi-View Conditioning”, *Array* (under review).
6. **Lahav, D.**, Ghosh, A., Schultz, M., Grosz, A., Klein, L., “Shape anisotropy in patterned soft ferromagnets is constant only above a crossover field”, *Communications Physics* (under review).
7. Vize, M., Alimi, R., Schultz, M., Fisher, E., **Lahav, D.**, Grosz, A., Klein, L., “Multi-Objective Optimization of Magnetic Anomaly Detection with Planar Hall Effect Sensors”, *IEEE Sensors Journal* (submitted).

CONFERENCE PRESENTATIONS

Oral Presentations

- “Multi-Functional Flexible Planar Hall Effect Sensors”, The 16th Joint Conference on Magnetism and Magnetic Materials and InterMag, 2025.
- “Multi-Functional Flexible Planar Hall Effect Sensors”, BINA Conference, Bar-Ilan University, 2025.
- “Flexible Spintronics Under Strain”, KLA-BINA Outstanding Students Seminar, Bar-Ilan University, 2026.
- “Multifunctional High-Resolution Flexible Elliptical Planar Hall Effect Sensors for Magnetometry and Strain Sensing”, XV European Magnetic Sensors and Actuators Conference (EMSA), 2026.
- “Enhanced Magnetic Resolution in Elliptical Planar Hall Effect Sensors via Non-Collinear Anisotropy Engineering”, Conference on Magnetism and Magnetic Materials (MMM), 2026 (accepted).
- “Multifunctional High-Resolution Flexible Elliptical Planar Hall Effect Sensors for Magnetometry and Strain Sensing”, Conference on Magnetism and Magnetic Materials (MMM), 2026 (accepted).

Poster Presentations

- “Planar Hall Effect Magnetic Sensors with Extended Field Range”, 69th Annual Meeting of the Israel Physical Society, 2024.
- “Expanding the Field Range of PHE Sensors for Increased Industrial Applicability”, International Symposium on Integrated Magnetism (iSIM), 2025.
- “Multi-Functional Flexible Planar Hall Effect Sensors”, International Symposium on Integrated Magnetism (iSIM), 2025.
- “Expanding the Field Range of PHE Sensors for Increased Industrial Applicability”, The 16th Joint Conference on Magnetism and Magnetic Materials and Intermag, 2025.
- “Expanding the Field Range of PHE Sensors for Increased Industrial Applicability”, BINA Conference, Bar-Ilan University, 2025.
- “Enhanced Magnetic Resolution in Elliptical Planar Hall Effect Sensors via Non-Collinear Anisotropy Engineering”, XV European Magnetic Sensors and Actuators Conference (EMSA), 2026.

FELLOWSHIPS, HONORS & AWARDS

M.Sc. in Physics - Magna Cum Laude	2024
Bar-Ilan University, Ramat-Gan, Israel.	
Excel@BINA Scholarship for Outstanding Students	2024 – 2025
Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.	
The President’s Scholarship Program for Outstanding Doctoral Fellows	2024 – 2025
Bar-Ilan University, Ramat-Gan, Israel.	
Excel@BINA Scholarship for Outstanding Students	2025 – 2026
Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.	
KLA Scholarship for Outstanding Students	2026
Awarded for a lecture at the KLA-BINA Outstanding Students Seminar, Institute of Nanotechnology and Advanced Materials, Bar-Ilan University.	

POSITIONS

Nano-Fabrication Center Operator, Bar-Ilan University, Ramat-Gan, Israel	2023 – Present
Support cleanroom nanofabrication and characterization workflows for interdisciplinary research projects:	
• Operate sputtering systems, lithography systems, and dicing equipment. • Perform thin-film deposition, lithography, and device fabrication. • Conduct AFM measurements in the MIXA characterization unit.	

TEACHING EXPERIENCE

Teaching Assistant, Bar-Ilan University, Ramat-Gan, Israel	2022 – Present
Teach undergraduate laboratory courses:	
• 86-251: Electronics Lab • 86-232: Computerized Physics Lab • 86-129: General Physics Lab • 96-900: Preparatory Physics Lab	
Tutor undergraduate course:	
• 86-250: Electronics for Physicists	

TECHNICAL SKILLS

Programming Languages: C++, MATLAB, Python

Software Tools: LabVIEW, COMSOL

Nanofabrication: Mask Aligner (MA), Maskless Aligner (MLA), Wafer Bonder, Ion-Beam Sputtering (IBS), BesTec (Sputtering and Evaporation), Disco Dicing Saw, ICP-RIE (Dry Etching)

Characterization: Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM)

CAD Tools: AutoCAD, Fusion 360, SolidWorks, CleWin

Systems Engineering: Design and integration of custom electrical, magnetic, and low-noise measurement systems, including automated experimental platforms

ACADEMIC SERVICE & OUTREACH

Student Representative, Physics Department – BIU Student Association 2023 – Present

Represent physics students in departmental and university committees and coordinate communication between students and faculty.

PRESS COVERAGE

“Flexible planar Hall effect sensor with sub-200 pT resolution”:

- [EurekAlert](#)
- [The Jerusalem Post](#)
- [AZO Sensors](#)
- [Israel National News](#)
- [Israel Noticias](#)